Introduction to Intelligent Agents

An intelligent agent perceives its environment through sensors and acts on that environment through actuators. A rational agent selects the action expected to maximize its performance measure, given the percept sequence and available knowledge. Rationality is not the same as omniscience: an agent cannot know the actual outcome of every action in advance.

The PEAS framework describes a task environment using Performance measure, Environment, Actuators, and Sensors. For a robot vacuum, a performance measure can include cleanliness and energy use. Its environment is the room. Wheels and suction are actuators, while dirt and distance detectors are sensors.

An environment is fully observable if sensors provide all information relevant to choosing an action. In a partially observable environment, the agent has incomplete information. An environment is deterministic if the next state is completely determined by the current state and the action taken. In a stochastic environment, outcomes involve uncertainty.

A simple reflex agent selects actions using condition-action rules based on the current percept. It does not retain percept history. A model-based reflex agent maintains internal state to track aspects of the environment that are not immediately observable. This internal state is updated using percepts and a model of how the world changes.

A goal-based agent chooses actions that move it toward a desired goal. It may use search and planning to consider future action sequences. A utility-based agent evaluates how desirable different outcomes are using a utility function. This helps compare alternatives that achieve a goal but differ in quality.

A learning agent improves with experience. Its learning element improves the performance element. A critic provides feedback on how well the agent is doing relative to a performance standard. A problem generator suggests exploratory actions that may produce new and informative experiences.

In a study assistant, lecture files form part of the environment. A controller coordinates the workflow. A planner breaks a student goal into steps. An executor performs the steps by calling tools. An observer checks whether a result is present and well-formed. A critic evaluates the result against lecture content. Memory stores session data and previous outputs.

A bounded revision loop limits the number of corrections. For this example, the assistant can revise an output at most once. A second review decides whether the corrected result is acceptable. This avoids indefinite retries, limits cost, and makes behavior easier to explain.

Grounding means basing an answer on the provided source material. Unsupported facts should be removed. A second AI review can help identify mistakes, but it does not guarantee that an answer is correct. Students should check important details against

their original lecture.